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Inventor(s)	Kim; Yangwan et al.

Display apparatus for preventing quality deterioration of displayed image

Abstract

A display apparatus includes: a thin-film transistor including a source electrode, a drain electrode, and a gate electrode; a data line in a layer different from the source electrode, the drain electrode, and the gate electrode, wherein the data line is configured to transmit a data signal; and a shield layer between the data line and a component of the thin-film transistor.

Inventors: Kim; Yangwan (Yongin-si, KR), Kwak; Wonkyu (Yongin-si, KR), Noh; Jaedu (Yongin-si, KR), Lee; Jaeyong (Yongin-si, KR)

Applicant: SAMSUNG DISPLAY CO., LTD. (Yongin-si, KR)

Family ID: 1000008820316

Assignee: Samsung Display Co., Ltd. (Yongin-si, KR)

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Primary Examiner: Whalen; Daniel

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/370,815, filed Jul. 8, 2021, which is a continuation of U.S. patent application Ser. No. 17/010,777, filed Sep. 2, 2020, now U.S. Pat. No. 11,063,107, which is a continuation of U.S. patent application Ser. No. 16/552,819, filed Aug. 27, 2019, now U.S. Pat. No. 10,784,329, which is a continuation of U.S. patent application Ser. No. 15/838,138, filed Dec. 11, 2017, now U.S. Pat. No. 10,439,015, which is a continuation of U.S. patent application Ser. No. 14/660,813, filed Mar. 17, 2015, now U.S. Pat. No. 9,842,892, which claims priority to and the

BACKGROUND

1. Field

(1) Embodiments of the present invention relate to a display apparatus.

2. Description of the Related Art

(2) In an organic light-emitting display apparatus, thin-film transistors (TFTs) may be located in each (sub) pixel to control the luminance of each (sub) pixel. Such TFTs control the luminance of the sub (pixel) according to a received data signal.

(3) However, luminance realized in a (sub) pixel of a general display apparatus may be different from that depending on a received data signal. Accordingly, an image displayed on the general display apparatus may have deteriorated quality.

SUMMARY

(4) One or more embodiments of the present invention include a display apparatus for preventing quality deterioration of a displayed image.

(5) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(6) According to some embodiments of the present invention, a display apparatus includes: a thin-film transistor including a source electrode, a drain electrode, and a gate electrode; a data line in a layer different from the source electrode, the drain electrode, and the gate electrode, wherein the data line is configured to transmit a data signal; and a shield layer between the data line and a component of the thin-film transistor.

(7) The shield layer may be between at least one of the source electrode and the data line, the drain electrode and the data line, or the gate electrode and the data line.

(8) The display apparatus may further include a storage capacitor including a first storage capacitor plate and a second storage capacitor plate, the second storage capacitor plate overlapping the first storage capacitor plate and being above the first storage capacitor plate, the data line may be in a layer above the second storage capacitor plate, the second storage capacitor plate may include the shield layer, and the shield layer may extend between the gate electrode and the data line or extends below the data line.

(9) The first storage capacitor plate may be electrically coupled to the gate electrode.

(10) The first storage capacitor plate and the gate electrode may be integrally formed.

(11) The gate electrode may include a first gate electrode and a second gate electrode, the shield layer may be between the data line and the component, and the component may be between the first gate electrode and the second gate electrode of the thin-film transistor.

(12) The display apparatus may further include a storage capacitor including a first storage capacitor plate and a second storage capacitor plate, the second storage capacitor plate overlapping the first storage capacitor plate and being above the first storage capacitor plate, the data line may be in a layer above the second storage capacitor plate, and the second storage capacitor plate may include the shield layer and extend at least one of between the data line and the component, between the first gate electrode and the second gate electrode of the thin-film transistor, or below the data line.

(13) The display apparatus may further include a driving thin-film transistor including a driving gate electrode electrically coupled to the first storage capacitor plate and a driving drain electrode electrically coupled to the source electrode, and the first storage capacitor plate may be electrically coupled to the drain electrode.

(14) The display apparatus may further include: a storage capacitor including a first storage capacitor plate and a second storage capacitor plate, the second capacitor plate overlapping the first storage capacitor plate and being above the first storage capacitor plate; and an initialization

voltage line configured to transmit an initialization voltage to a driving gate electrode electrically coupled to the first storage capacitor plate of a driving thin-film transistor, the initialization voltage line may be in a same layer as the second storage capacitor plate, the drain electrode may be electrically coupled to the first storage capacitor plate and the source electrode may be electrically coupled to the initialization voltage line, the data line may be in a layer above the second storage capacitor plate, and the shield layer may be integrated with the initialization voltage line and may extend at least one of between the data line and the component, above the component between the first gate electrode and the second gate electrode of the thin-film transistor, or below the data line.

(15) According to some embodiments of the present invention, a display apparatus includes: a thin-film transistor including a source electrode, a drain electrode, and a gate electrode; a data line in a layer different from the source electrode, the drain electrode, and the gate electrode, wherein the data line is configured to transmit a data signal; a storage capacitor including: a first storage capacitor plate electrically coupled to the drain electrode; and a second storage capacitor plate in a layer different from the first storage capacitor plate, wherein the second storage capacitor plate overlaps the first storage capacitor plate; and an initialization voltage line configured to transmit an initialization voltage to a driving gate electrode electrically coupled to the first storage capacitor plate of a driving thin-film transistor and is electrically coupled to the source electrode, wherein the gate electrode includes a first gate electrode and a second gate electrode, and one of the first gate electrode and the second gate electrode is at least partially between the data line and a component between the first gate electrode and the second gate electrode of the thin-film transistor.

(16) The one of the first gate electrode and the second gate electrode may extend below or above the data line.

(17) According to some embodiments of the present invention, a display apparatus includes: a thin-film transistor including a source electrode, a drain electrode, and a gate electrode; a control signal line in a layer different from the source electrode, the drain electrode, and the gate electrode, and wherein the control signal line is configured to transmit a control signal; and a shield layer between the control signal line and a component of the thin-film transistor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1 is a circuit diagram of a (sub) pixel of an organic light-emitting display apparatus, according to an embodiment of the present invention;
- (3) FIG. 2 is a schematic diagram showing locations of a plurality of thin-film transistors (TFTs) and a capacitor in the (sub) pixel of FIG. 1, according to an embodiment of the present invention;
- (4) FIGS. 3 through 6 are schematic diagrams showing components of the plurality of TFTs and the capacitor of FIG. 2 by each layer;
- (5) FIG. 7 is a cross-sectional view taken along the line VII-VII of FIG. 2;
- (6) FIG. 8 is a cross-sectional view of an organic light-emitting display apparatus according to some embodiments of the present invention;
- (7) FIG. 9 is a cross-sectional view taken along the line IX-IX of FIG. 2;
- (8) FIG. 10 is a cross-sectional view taken along the line X-X of FIG. 2;
- (9) FIG. 11 is a schematic diagram showing locations of a plurality of TFTs and a capacitor in a (sub) pixel of an organic light-emitting display apparatus, according to another embodiment of the present invention; and
- (10) FIG. 12 is a cross-sectional view taken along the line XII-XII of FIG. 11.

DETAILED DESCRIPTION

(11) Aspects and features of one or more embodiments of the present invention and methods of accomplishing the same may be understood more readily by reference to the following detailed description of the embodiments and the accompanying drawings. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Rather, these embodiments are provided so that this disclosure will be more thorough and complete and will more fully convey the concepts of the present embodiments to one of ordinary skill in the art, and the present invention will only be defined by the appended claims and their equivalents.

(12) Hereinafter, one or more embodiments of the present invention will be described below in more detail with reference to the accompanying drawings. Those components that are the same or are in correspondence are rendered the same reference numeral regardless of the figure number, and redundant explanations are omitted.

(13) It will be understood that when a layer, region, or component is referred to as being “formed on,” another layer, region, or component, it can be directly or indirectly formed on the other layer, region, or component. That is, for example, intervening layers, regions, or components may be present. Sizes of elements in the drawings may be exaggerated for convenience of explanation. In other words, since sizes and thicknesses of components in the drawings are arbitrarily illustrated for convenience of explanation, the following embodiments are not limited thereto.

(14) In the following examples, the x-axis, the y-axis and the z-axis are not limited to three axes of the rectangular coordinate system, and may be interpreted in a broader sense. For example, the x-axis, the y-axis, and the z-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another.

(15) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(16) FIG. 1 is an equivalent circuit diagram of a (sub) pixel of an organic light-emitting display apparatus, according to some embodiments of the present invention.

(17) As shown in FIG. 1, the (sub) pixel of the organic light-emitting display apparatus according to an embodiment includes a plurality of signal lines, a plurality of thin-film transistors (TFTs) coupled to the plurality of signal lines, a storage capacitor Cst, and an organic light-emitting device OLED. Here, the plurality of signal lines may be shared by a plurality of (sub) pixels.

(18) The plurality of TFTs include a driving TFT T1, a switching TFT T2, a compensating TFT T3, an initialization TFT T4, an operation control TFT T5, and an emission control TFT T6.

(19) The plurality of signal lines include a scan line 121 transmitting a scan signal Sn, a previous scan line 122 transmitting a previous scan line Sn-1 to the initialization TFT T4, an emission control line 123 transmitting an emission control signal En to the operation control TFT T5 and the emission control TFT T6, a data line 171 crossing the scan line 121 and transmitting a data signal Dm, a driving voltage line 172 transmitting a driving voltage ELVDD and disposed almost in parallel to the data line 171, and an initialization voltage line 124 transmitting an initialization voltage Vint for initializing the driving TFT T1.

(20) A gate electrode G1 of the driving TFT T1 is coupled to a first storage capacitor plate Cst1 of the storage capacitor Cst, a source electrode S1 of the driving TFT T1 is coupled to the driving voltage line 172 through the operation control TFT T5, and a drain electrode D1 of the driving TFT T1 is electrically coupled to a pixel electrode of the organic light-emitting device OLED through the emission control TFT T6. According to a switching operation of the switching TFT T2, the driving TFT T1 receives the data signal Dm and supplies a driving current I.sub.OLED to the organic light-emitting device OLED.

(21) A gate electrode G2 of the switching TFT T2 is coupled to the scan line 121, a source electrode S2 of the switching TFT T2 is coupled to the data line 171, and a drain electrode D2 of the switching TFT T2 is coupled to the source electrode S1 of the driving TFT T1 and coupled to

the driving voltage line **172** through the operation control TFT T5. Such a switching TFT T2 is turned on according to the scan signal Sn received through the scan line **121**, and performs a switching operation by transmitting the data signal Dm from the data line **171** to the source electrode S1 of the driving TFT T1.

(22) A gate electrode G3 of the compensation TFT T3 is coupled to the scan line **121**, a source electrode S3 of the compensating TFT T3 is coupled to the drain electrode D1 of the driving TFT T1 while being coupled to the pixel electrode of the organic light-emitting device OLED through the emission control TFT T6, and a drain electrode D3 of the compensating TFT T3 is coupled to the first storage capacitor plate Cst1 of the storage capacitor Cst, a drain electrode D4 of the initialization TFT T4, and the gate electrode G1 of the driving TFT T1. Such a compensating TFT T3 is turned on according to the scan signal Sn received through the scan line **121**, and diode-connects the driving TFT T1 by electrically coupling the gate electrode G1 and the drain electrode D1 of the driving TFT T1.

(23) A gate electrode G4 of the initialization TFT T4 is coupled to the previous scan line **122**, a source electrode S4 of the initialization TFT T4 is coupled to the initialization voltage line **124**, and the drain electrode D4 of the initialization TFT T4 is coupled to the first storage capacitor plate Cst1 of the storage capacitor Cst, the drain electrode D3 of the compensating TFT T3, and the gate electrode G1 of the driving TFT T1. The initialization TFT T4 is turned on according to the previous scan signal Sn-1 received through the previous scan line **122**, and performs an initialization operation by initializing a voltage of the gate electrode G1 of the driving TFT T1 by transmitting the initialization voltage Vint to the gate electrode G1 of the driving TFT T1.

(24) A gate electrode G5 of the operation control TFT T5 is coupled to the emission control line **123**, a source electrode S5 of the operation control TFT T5 is coupled to the driving voltage line **172**, and a drain electrode D5 of the operation control TFT T5 is coupled to the source electrode S1 of the driving TFT T1 and the drain electrode D2 of the switching TFT T2.

(25) A gate electrode G6 of the emission control TFT T6 is coupled to the emission control line **123**, a source electrode S6 of the emission control TFT T6 is coupled to the drain electrode D1 of the driving TFT T1 and the source electrode S3 of the compensating TFT T3, and a drain electrode D6 of the emission control TFT T6 is electrically coupled to the pixel electrode of the organic light-emitting device OLED. The operation control TFT T5 and an emission control TFT T6 are concurrently (e.g., simultaneously) turned on according to the emission control signal En received through the emission control line **123**, and transmit the driving voltage ELVDD to the organic light-emitting device OLED such that the driving current I.sub.OLED flows through the organic light-emitting device OLED.

(26) A second storage capacitor plate Cst2 of the storage capacitor Cst is coupled to the driving voltage line **172**, and a counter electrode of the organic light-emitting device OLED is coupled to a common voltage ELVSS. Accordingly, the organic light-emitting device OLED emits light by receiving the driving current I.sub.OLED from the driving TFT T1, thereby displaying an image.

(27) Detailed operations of a pixel in such an organic light-emitting display apparatus will now be briefly described.

(28) First, the previous scan signal Sn-1 in a low level is supplied through the previous scan line **122** during an initialization period. Then, the initialization TFT T4 is turned on in response to the previous scan signal Sn-1 in the low level, and thus the initialization voltage Vint is transmitted to the gate electrode G1 of the driving TFT T1 from the initialization voltage line **124** through the initialization TFT T4, and the driving TFT T1 is initialized by the initialization voltage Vint.

(29) Then, the scan signal Sn in a low level is supplied through the scan line **121** during a data programming period. Accordingly, the switching TFT T1 and the compensating TFT T3 are turned on in response to the scan signal Sn in the low level. Thus, the driving TFT T1 is diode-coupled by the turned on compensating TFT T3 and is biased in a forward direction. Then, a compensating voltage ($Dm + V_{th}$, wherein V_{th} has a negative value) obtained by subtracting a threshold voltage

V_{th} of the driving TFT T1 from the data signal D_m supplied from the data line 171 is applied to the gate electrode G1. Next, the driving voltage ELVDD and the compensating voltage are applied to two ends of the storage capacitor C_{st}, and thus a charge corresponding to a voltage difference between the two ends is stored in the storage capacitor C_{st}.

(30) Then, the emission control signal E_n supplied from the emission control line 123 during an emission period is changed from a high level to a low level. Accordingly, the operation control TFT T5 and the emission control TFT T6 are turned on according to the emission control signal E_n in the low level during the emission period. Then, the driving current I_{sub.OLED} determined based on a voltage difference between a voltage of the gate electrode G1 of the driving TFT T1 and the driving voltage ELVDD is generated, and the driving current I_{sub.OLED} is supplied to the organic light-emitting device OLED through the emission control TFT T6. A gate-source voltage V_{sub.GS} of the driving TFT T1 maintains '(D_m+V_{th})-ELVDD' by the storage capacitor C_{st} during the emission period, and because the driving current I_{sub.OLED} is proportional to '(D_m-ELVDD).sup.2' (i.e., a square of a value obtained by subtracting the threshold voltage V_{th} from the gate-source voltage V_{sub.GS}, according to a current-voltage relationship of the driving TFT T1), the driving current I_{sub.OLED} may be determined regardless of the threshold voltage V_{th} of the driving TFT T1.

(31) A more detailed structure of the (sub) pixel of the organic light-emitting display apparatus of FIG. 1 will now be described with reference to FIGS. 2 through 10.

(32) FIG. 2 is a schematic diagram showing locations of the plurality of TFTs and the capacitor in the (sub) pixel of FIG. 1, according to an embodiment of the present invention. FIGS. 3 through 6 are schematic diagrams showing components of the plurality of TFTs and the capacitor of FIG. 2 by each layer. In other words, each of FIGS. 3 through 6 illustrates an arrangement of a wire or a semiconductor layer disposed in the same layer, and an insulating layer may be located between layers shown in FIGS. 3 through 6. For example, a first insulating layer 141 of FIG. 7 may be located between the layer of FIG. 3 and the layer of FIG. 4, a second insulating layer 142 of FIG. 7 may be located between the layer of FIG. 4 and the layer of FIG. 5, and an interlayer insulating layer 160 of FIG. 7 may be located between the layer of FIG. 5 and the layer of FIG. 6. Here, contact holes etc. may be formed on such insulating layers so that the layers of FIGS. 3 through 6 are electrically coupled to each other.

(33) The (sub) pixel of the organic light-emitting display apparatus, according to the current embodiment, includes the scan line 121, the previous scan line 122, the emission control line 123, and the initialization voltage line 124, which respectively apply the scan signal S_n, the previous scan signal S_{n-1}, the emission control signal E_n, and the initialization voltage V_{int} and are formed along a row direction. Also, the (sub) pixel of the organic light-emitting display apparatus, according to the current embodiment, may include the data line 171 and the driving voltage line 172, which cross the scan line 121, the previous scan line 122, the emission control line 123, and the initialization voltage line 124, and respectively apply the data signal D_m and the driving voltage ELVDD to the (sub) pixel.

(34) Also, the (sub) pixel may include the driving TFT T1, the switching TFT T2, the compensating TFT T3, the initialization TFT T4, the operation control TFT T5, the emission control TFT T6, the storage capacitor C_{st}, and an organic light-emitting device.

(35) The driving TFT T1, the switching TFT T2, the compensating TFT T3, the initialization TFT T4, the operation control TFT T5, and the emission control TFT T6 are formed on a semiconductor layer as shown in FIG. 3, wherein the semiconductor layer may have a shape curving or contouring in any shape. The semiconductor layer may include a driving semiconductor layer 131a corresponding to the driving TFT T1, a switching semiconductor layer 131b corresponding to the switching TFT T2, a compensating semiconductor layer 131c1, 131c2, and 131c3 corresponding to the compensating TFT T3, initialization semiconductor layer 131d1, 131d2, and 131d3 corresponding to the initialization TFT T4, an operation control semiconductor layer 131e

corresponding to the operation control TFT T5, and an emission control semiconductor layer **131f** corresponding to the emission control TFT T6. In other words, the driving semiconductor layer **131a**, the switching semiconductor layer **131b**, the compensating semiconductor layer **131c1**, **131c2**, and **131c3**, the initialization semiconductor layer **131d1** through **131d3**, the operation control semiconductor layer **131e**, and the emission control semiconductor layer **131f** may be understood to constitute partial regions of the semiconductor layer of FIG. 3.

(36) The semiconductor layer may include polysilicon. Also, the semiconductor layer may include, for example, a channel region that is not doped with an impurity, and source and drain regions that are formed as impurities are doped on two sides of the channel region. Here, the impurities may vary according to a type of a TFT, and may be N-type impurities or P-type impurities. Also, the source or drain region may be interpreted as a source or drain electrode of a TFT. In other words, for example, a driving source electrode **176a** may correspond to a driving drain region doped with an impurity near the driving semiconductor layer **131a** of the semiconductor layer of FIG. 3, and a driving drain electrode **177a** may correspond to a driving drain region doped with an impurity near the driving semiconductor layer **131a** of the semiconductor layer of FIG. 3. Also, a region of the semiconductor layer of FIG. 3 between TFTs may be doped with an impurity to operate as a wire electrically coupling the TFTs.

(37) Meanwhile, the storage capacitor Cst may be formed. The storage capacitor Cst may include a first storage capacitor plate **125a** and a second storage capacitor plate **127**, wherein the second insulating layer **142** is located therebetween. Here, the first storage capacitor plate **125a** may also operate as a driving gate electrode of the driving TFT T1. In other words, the driving gate electrode and the first storage capacitor plate **125a** may be integrally formed. Hereinafter, for convenience, the same reference numeral as the first storage capacitor plate **125a** may be used for the driving gate electrode.

(38) The first storage capacitor plate **125a** may have a rectangular shape isolated from an adjacent (sub) pixel as shown in FIG. 4. Such a first storage capacitor plate **125a** may be formed in the same layer and of the same material as the scan line **121**, the previous scan line **122**, and the emission control line **123** as shown in FIG. 4.

(39) For reference, a switching gate electrode **125b** and a compensating gate electrode **125c1** and **125c2** may be a part of the scan line **121** crossing the semiconductor layer or a part protruding from the scan line **121**, an initialization gate electrode **125d1** and **125d2** may be parts of the previous scan line **122** crossing the semiconductor layer or parts protruding from the previous scan line **122**, and an operation control gate electrode **125e** and an emission control gate electrode **125f** may be parts of the emission control line **123** crossing the semiconductor layer or parts protruding from the emission control line **123**.

(40) The second storage capacitor plates **127** of adjacent (sub) pixels may be coupled to each other, and as shown in FIG. 5, may be formed in the same layer and of the same material as the initialization voltage line **124**. A storage opening **27** may be formed on the second storage capacitor plate **127** and may enable the first storage capacitor plate **125a** and a compensating drain electrode **177c** of the compensating TFT T3 to be electrically coupled to each other through a connecting unit **174** described in more detail later. The second storage capacitor plate **127** may be coupled to the driving voltage line **172** through a contact hole **168** formed on the interlayer insulating layer **160**.

(41) The driving TFT T1 includes the driving semiconductor layer **131a**, the driving gate electrode **125a**, the driving source electrode **176a**, and the driving drain electrode **177a**. As described above, the driving gate electrode **125a** may also operate as the first storage capacitor plate **125a**. The driving source electrode **176a** is an external region (in -x direction in FIG. 3) of the driving gate electrode **125a**, and the driving drain electrode **177a** is an external region (in +x direction in FIG. 3) of the driving gate electrode **125a** and is arranged opposite to the driving source electrode **176a** based on the driving gate electrode **125a**.

(42) The switching TFT T2 includes the switching semiconductor layer **131b**, the switching gate

electrode **125b**, a switching source electrode **176b**, and a switching drain electrode **177b**. The switching source electrode **176b** may be electrically coupled to the data line **171** through a contact hole **164** formed through the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**. Here, if required, a part near the contact hole **164** of the data line **171** may be understood to be the source electrode S2 of the switching TFT T2. The switching drain electrode **177b** corresponds to a switching drain region doped with an impurity near the switching semiconductor layer **131b**.

(43) The compensating TFT T3 includes the compensating semiconductor layer **131c1**, **131c2**, and **131c3**, the compensating gate electrode **125c1** and **125c2**, a compensating source electrode **176c**, and the compensating drain electrode **177c**. The compensating source electrode **176c** corresponds to a compensating source region doped with an impurity near the compensating semiconductor layer, and the compensating drain electrode **177c** corresponds to a compensating drain region doped with an impurity near the compensating semiconductor layer. The compensating gate electrode **125c1** and **125c2** is a dual gate electrode including a first gate electrode **125c1** and a second gate electrode **125c2**, and may prevent or reduce generation of a leakage current. The compensating drain electrode **177c** of the compensating TFT T3 may be coupled to the first storage capacitor plate **125a** through the connecting unit **174**. The compensating semiconductor layer may include a part or component **131c1** corresponding to the first gate electrode **125c1**, a part or component **131c3** corresponding to the second gate electrode **125c2**, and a part or component **131c2** between the parts **131c1** and **131c3**.

(44) As shown in FIG. 6, the connecting unit **174** may be formed of the same material and in the same layer as the data line **171**. One end of the connecting unit **174** is coupled to the compensating drain electrode **177c** and an initialization drain electrode **177d** through a contact hole **166** formed through the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**, and the other end of the connecting unit **174** is coupled to the first storage capacitor plate **125a** through a contact hole **167** formed through the second insulating layer **142** and the interlayer insulating layer **160**. Here, the other end of the connecting unit **174** is coupled to the first storage capacitor plate **125a** through the storage opening **27** formed on the second storage capacitor plate **127**.

(45) The initialization TFT T4 includes an initialization semiconductor layer **131d1**, **131d2**, and **131d3**, an initialization gate electrode **125d1** and **125d2**, an initialization source electrode **176d**, and the initialization drain electrode **177d**. The initialization drain electrode **177d** corresponds to an initialization drain region doped with an impurity near the initialization semiconductor layer **131d1**, **131d2**, and **131d3**.

(46) The initialization source electrode **176d** is coupled to the initialization voltage line **124** through an initialization connecting line **78**. One end of the initialization connecting line **78** may be coupled to the initialization voltage line **124** through a contact hole **161** formed through the second insulating layer **142** and the interlayer insulating layer **160**, and the other end of the initialization connecting line **78** may be coupled to the initialization source electrode **176d** through a contact hole **162** formed through the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**.

(47) The operation control TFT T5 includes the operation control semiconductor layer **131e**, the operation control gate electrode **125e**, an operation control source electrode **176e**, and an operation control drain electrode **177e**. The operation control source electrode **176e** may be electrically coupled to the driving voltage line **172** through a contact hole **165** formed through the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**. Here, if required, a part near the contact hole **165** of the driving voltage line **172** may be understood to be the source electrode S5 of the operation control TFT T5. The operation control drain electrode **177e** corresponds to an operation control drain region doped with an impurity near the operation control semiconductor layer **131e**.

(48) The emission control TFT T6 includes the emission control semiconductor layer **131f**, the emission control gate electrode **125f**, an emission control source electrode **176f**, and an emission control drain electrode **177f**. The emission control source electrode **176f** corresponds to an emission control source region doped with an impurity near the emission control semiconductor layer **131f**. As shown in FIG. 6, the emission control drain electrode **177f** may be understood to be a part formed on the interlayer insulating layer **160** together with the data line **171** or the driving voltage line **172**. The emission control drain electrode **177f** may be coupled to a lower semiconductor layer through a contact hole **163** formed through the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**. Alternatively, it may be understood that a part of the lower semiconductor layer is a emission control drain electrode, and the reference numeral **177f** denote an intermediate connection layer for coupling the emission control drain electrode and the pixel electrode of the organic light-emitting device OLED.

(49) One end of the driving semiconductor layer **131a** of the driving TFT T1 is coupled to the switching semiconductor layer **131b** and the compensating semiconductor layer, and the other end of the driving semiconductor layer **131a** is coupled to the operation control semiconductor layer **131e** and the emission control semiconductor layer **131f**. Accordingly, the driving source electrode **176a** is coupled to the switching drain electrode **177b** and the operation control drain electrode **177e**, and the driving drain electrode **177a** is coupled to the compensating source electrode **176c** and the emission control source electrode **176f**.

(50) Meanwhile, the switching TFT T2 is used as a switching device for selecting a (sub) pixel to emit light. The switching gate electrode **125b** is coupled to the scan line **121**, the switching source electrode **176b** is coupled to the data line **171**, and the switching drain electrode **177b** is coupled to the driving TFT T1 and the operation control TFT T5.

(51) Also, as shown in FIG. 6, the emission control drain electrode **177f** of the emission control TFT T6 is coupled to the pixel electrode of the organic light-emitting device OLED through a contact hole **181** formed on a passivation film or planarization film covering the data line **171** or the driving voltage line **172** formed in the same layer.

(52) FIG. 7 is a cross-sectional view taken along the line VII-VII of FIG. 2. As shown in FIG. 7, various components described above may be arranged on a substrate **110**. The substrate **110** may be formed of any one of various suitable substrate materials, such as glass, a metal, and plastic. A buffer layer **111** may be disposed on the substrate **110** as occasion demands. The buffer layer **111** may flatten a surface of the substrate **110** or prevent impurities from penetrating into a semiconductor layer on the substrate **110**. Such a buffer layer **111** may be a single layer or multilayer structure formed of silicon oxide, silicon nitride, or silicon oxynitride.

(53) The driving semiconductor layer **131a**, the switching semiconductor layer **131b**, and the compensating semiconductor layer described above with reference to FIG. 3 may be arranged on the buffer layer **111**. The first insulating layer **141** formed of silicon nitride, silicon oxide, or silicon oxynitride may be arranged on the driving semiconductor layer **131a**, the switching semiconductor layer **131b**, and the compensating semiconductor layer.

(54) Wires including the driving gate electrode **125a**, the scan line **121** including the switching gate electrode **125b** and the compensating gate electrode **125c1** and **125c2**, the previous scan line **122** including the initialization gate electrode **125d1** and **125d2**, the emission control line **123** including the operation control gate electrode **125e** and the emission control gate electrode **125f**, which have been described above with reference to FIG. 4, may be arranged on the first insulating layer **141**. The driving gate electrode **125a**, the scan line **121**, the previous scan line **122**, and the emission control line **123** may be collectively referred to as a first gate wire.

(55) The second insulating layer **142** may cover the first gate wire. The second insulating layer **142** may be formed of silicon nitride, silicon oxide, or silicon oxynitride. The second storage capacitor plate **127** and the initialization voltage line **124** described above with reference to FIG. 5 may be arranged on the second insulating layer **142**. The second storage capacitor plate **127** and the

initialization voltage line **124** may be collectively referred to as a second gate wire.

(56) The interlayer insulating layer **160** is disposed on the second gate wire. The interlayer insulating layer **160** may be formed of silicon nitride, silicon oxide, or silicon oxynitride.

(57) The data line **171**, the driving voltage line **172**, the connecting unit **174**, the initialization connecting line **78**, and the emission control drain electrode **177f**, which have been described above with reference to FIG. **6**, may be arranged on the interlayer insulating layer **160**. The data line **171**, the driving voltage line **172**, the connecting unit **174**, the initialization connecting line **78**, and the emission control drain electrode **177f** may be collectively referred to as a data wire. As described above, the data line **171**, the driving voltage line **172**, the connecting unit **174**, the initialization connecting line **78**, and the emission control drain electrode **177f** may be electrically coupled to a lower semiconductor layer or an electrode through the contact holes **161** through **168** formed on at least a part of the first insulating layer **141**, the second insulating layer **142**, and the interlayer insulating layer **160**.

(58) A passivation film or planarization film is disposed on the data wire, and a pixel electrode of an organic light-emitting device may be arranged on the passivation film or planarization film. The pixel electrode may be coupled to the emission control drain electrode **177f** through the contact hole **181** formed on the passivation film or planarization film.

(59) Meanwhile, as shown in FIGS. **2**, **5**, and **7**, the second storage capacitor plate **127** may include a first shield layer SD1 at one side. As shown in FIGS. **2** and **5**, the first shield layer SD1 may be a part protruding from the second storage capacitor plate **127**. The first shield layer SD1 may be understood as a part of the second storage capacitor plate **127**, which extends between the data line **171** and at least a part between the first gate electrode **125c1** and the second gate electrode **125c2** of the compensating TFT T3.

(60) For reference, FIG. **2** is a diagram of one (sub) pixel, and a (sub) pixel having the same or similar structure may be disposed top, bottom, left, and right of the (sub) pixel. In FIG. **7**, a (sub) pixel P1 corresponds to the (sub) pixel of FIG. **2**, and a part of a (sub) pixel P2 disposed next to the (sub) pixel P1 in the +x direction of FIG. **2** are illustrated. The (sub) pixel P2 may also include the data line **171**, and accordingly, the first shield layer SD1 of the (sub) pixel P1 may be understood to be a part of the second storage capacitor plate **127**, which extends between the data line **171** of the (sub) pixel P2 and at least a part between the first and second gate electrodes **125c1** and **125c2** of the compensating TFT T3.

(61) If the first shield layer SD1 does not exist, the components between the first and second gate electrodes **125c1** and **125c2** of the compensating TFT T3, for example, the part **131c2** of the compensating semiconductor layer, may be affected by the data line **171**.

(62) The data line **171** transmits a data signal to the (sub) pixel P2 disposed near the (sub) pixel P1 in the +x direction, and also transmits a data signal to a plurality of (sub) pixels disposed near the (sub) pixel P2 in +y and -y directions. Here, a data signal being transmitted may vary according to luminance to be realized in the plurality of (sub) pixels disposed near the (sub) pixel P2 in the +y and -y directions, and accordingly, the data line **171** near the part **131c2** of the compensating semiconductor layer of the (sub) pixel P1 may transmit different electric signals according to time while the (sub) pixel P1 emits light.

(63) If the first shield layer SD1 does not exist, parasitic capacitance may occur between the data line **171** of the (sub) pixel P2 and the part **131c2** of the compensating TFT T3 of the (sub) pixel P1, and accordingly, the electric potential of the part **131c2** of the compensating TFT T3 of the (sub) pixel P1 may be affected by different electric signals transmitted by the data line **171** of the (sub) pixel P2 according to time while the (sub) pixel P1 emits light. Because the compensating TFT T3 is electrically coupled to the driving TFT T1, if the electric potential of the part **131c2** of the compensating TFT T3 of the (sub) pixel P1 is affected by the different electric signals transmitted by the data line **171** of the (sub) pixel P2, the luminance of the organic light-emitting device OLED determined by the driving TFT T1 may become different from an initial intensity, and thus quality

of an image displayed by the organic light-emitting display apparatus may deteriorate.

(64) However, according to the organic light-emitting display apparatus of some embodiments, because the first shield layer SD1 is disposed between the data line 171 of the (sub) pixel P2 and the part 131c2 of the compensating TFT T3 of the (sub) pixel P1, the part 131c2 of the compensating TFT T3 of the (sub) pixel P1 may not be affected or may be less affected by the data line 171 of the (sub) pixel P2, and thus the organic light-emitting display apparatus may be able to display an image having a more accurate luminance and a relatively higher quality. For example, if the first shield layer SD1 is a part of the second storage capacitor plate 127, the second storage capacitor plate 127 is connected to the driving voltage line 172 always having uniform electric potential, through the contact hole 168, and thus the first shield layer SD1 may also always have a uniform electric potential. Accordingly, an effect of an adjacent electric signal on the part 131c2 of the compensating TFT T3 may be reduced.

(65) Alternatively, the first shield layer SD1 may extend below the data line 171 of the (sub) pixel P2 as shown in FIG. 8 that is a cross-sectional view of an organic light-emitting display apparatus according to some embodiments of the present invention. Accordingly, the part 131c2 of the compensating TFT T3 may be further shielded. Here, the part 131c2 of the compensating TFT T3 may also be further shielded by extending the first shield layer SD1 above at least a part of the part 131c2 between the first and second gate electrodes 125c1 and 125c2 of the compensating TFT T3.

(66) FIG. 9 is a cross-sectional view taken along the line IX-IX of FIG. 2. As shown in FIGS. 2, 5, and 9, the initialization voltage line 124 may include a second shield layer SD2.

(67) As shown in FIGS. 2 and 5, the second shield layer SD2 may be a part of the initialization voltage line 124, which extends along an x-axis. The second shield layer SD2 may be understood to be a part of the initialization voltage line 124, which extends between the data line 171 and at least a part between first gate electrode 125d1 and a second gate electrode 125d2 of the initialization gate electrode 125d1 and 125d2 of the initialization TFT T4.

(68) In FIGS. 2 and 5, the initialization voltage line 124 extends above the part between the first and second gate electrodes 125d1 and 125d2 of the initialization TFT T4, but an embodiment is not limited thereto. If the initialization voltage line 124 has another location or another shape, for example, is moved in a +y direction, a -y direction, or another direction, or is curved, the initialization voltage line 124 may have a protrusion and the protrusion may extend between the data line 171 and at least a part between the first and second gate electrodes 125d1 and 125d2 of the initialization TFT T4 to operate as the second shield layer SD2. In other words, in FIGS. 2 and 5, the initialization voltage line 124 may extend along an x-axis direction while passing between the data line 171 and at least the part between the first and second gate electrodes 125d1 and 125d2 of the initialization TFT T4, such that a location of the initialization voltage line 124 is specified without having to include the protrusion.

(69) If the second shield layer SD2 does not exist, the components between the first and second gate electrodes 125d1 and 125d2 of the initialization TFT T4, for example, a part 131d2 of the initialization semiconductor layer 131d1, 131d2, and 131d3 may be affected by the data line 171.

(70) The data line 171 transmits a data signal to the (sub) pixel of FIG. 2, and also transmits a data signal to a plurality of (sub) pixels disposed near the (sub) pixel of FIG. 2 in +y and -y directions. Here, a data signal being transmitted may vary according to luminance to be realized by the plurality of (sub) pixels disposed near the (sub) pixel of FIG. 2 in the +y and -y directions, and accordingly, the data line 171 near the part 131d2 of the initialization semiconductor layer 131d1, 131d2, and 131d3 of the (sub) pixel of FIG. 2 transmits different electric signals according to time while the (sub) pixel of FIG. 2 emits light.

(71) If the second shield layer SD2 does not exist, parasitic capacitance may occur between the data line 171 and the part 131d2 of the initialization semiconductor layer 131d1, 131d2, and 131d3 of the initialization TFT T4, and thus electric potential of the part 131d2 of the initialization semiconductor layer 131d1, 131d2, and 131d3 of the initialization TFT T4 may be affected by the

different electric signals transmitted by the data line **171**, according to time while the (sub) pixel of FIG. **2** emits light. Because the initialization TFT **T4** is electrically coupled to the driving TFT **T1**, if the electric potential of the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** of the initialization TFT **4** is affected by the different electric signals transmitted by the data line **171**, the luminance of the organic light-emitting device OLED determined by the driving TFT **T1** may become different from an initial intention, and thus quality of an image displayed by the organic light-emitting display apparatus may deteriorate.

(72) However, according to the organic light-emitting display apparatus of some embodiments, because the second shield layer **SD2** is arranged between the data line **171** and the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** of the initialization TFT **T4**, the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** of the initialization TFT **T4** may not be affected or may be less affected by the data line **171**, and thus the organic light-emitting display apparatus may be able to display an image having a more accurate luminance and a relatively higher quality. For example, if the second shield layer **SD2** is a part of the initialization voltage line **124**, the second shield layer **SD2** may always have a uniform electric potential by the initialization voltage line **124** that always has uniform electric potential. Accordingly, an effect of an adjacent electric signal on the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** of the initialization TFT **T4** may be reduced.

(73) Here, if a layout of various wires or a semiconductor layer differs from that shown in FIG. **2**, the second shield layer **SD2** may be a part extending at least above the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** between the first and second gate electrodes **125d1** and **125d2** of the initialization TFT **T4**, or a part extending below the data line **171**.

(74) FIG. **10** is a cross-sectional view taken along the line X-X of FIG. **2**. As shown in FIGS. **2**, **5**, and **10**, the second storage capacitor plate **127** may include a third shield layer **SD3**.

(75) As shown in FIGS. **2** and **5**, the third shield layer **SD3** may be a part of the second storage capacitor plate **127**. The third shield layer **SD3** may be understood to be a part of the second storage capacitor plate **127**, which extends between the data line **171** and the driving gate electrode **125a** of the driving TFT **T1**. For example, the second storage capacitor plate **127** may have a (virtual) end in the $-x$ direction, which approximately match an end of the first storage capacitor plate **125a** in the $-x$ direction below the second storage capacitor plate **127**, and the third shield layer **SD3** may exist between the data line **171** and the driving gate electrode **125a** of the driving TFT **T1** in the $-x$ direction from the (virtual) end, wherein the third shield layer **SD3** may be understood to be integrally formed with the second storage capacitor plate **127**.

(76) Alternatively, unlike shown in FIGS. **2** and **5**, the second storage capacitor plate **127** may not extend in the $-x$ direction where the data line **171** is located due to nonexistence of the third shield layer **SD3**, and the end of the second storage capacitor plate **127** in the $-x$ direction may approximately match the end of the first storage capacitor plate **125a** in the $-x$ direction. In this case, the driving gate electrode **125a** of the driving TFT **T1** is affected by the data line **171**.

(77) The data line **171** transmits a data signal to the (sub) pixel of FIG. **2**, and also transmits a data signal to the plurality of (sub) pixels disposed in the $+y$ and $-y$ directions of the (sub) pixel. Here, the data signal being transmitted may vary according to the luminance to be realized by the plurality of (sub) pixels disposed in the $+y$ and $-y$ directions of the (sub) pixel of FIG. **2**, and accordingly, the data line **171** near the part **131d2** of the initialization semiconductor layer **131d1**, **131d2**, and **131d3** of the (sub) pixel of FIG. **2** may transmit different electric signals according to time while the (sub) pixel of FIG. **2** emits light.

(78) If the third shield layer **SD3** does not exist and thus the second storage capacitor plate **127** does not extend in the $-x$ direction where the data line **171** is disposed and the end of the second storage capacitor plate **127** in the $-x$ direction approximately matches the end of the first storage capacitor plate **125a** in the $-x$ direction, parasitic capacitance exists between the data line **171** and

the driving gate electrode **125a** of the driving TFT **T1**, and accordingly, electric potential of the driving gate electrode **125a** of the driving TFT **T1** is affected by the different electric signals transmitted by the data line **171** according to time while the (sub) pixel of FIG. **2** emits light. As a result, the luminance of the organic light-emitting device OLED determined by the driving TFT **T1** may become different from an initial intension, and thus quality of an image displayed by the organic light-emitting display apparatus may deteriorate.

(79) However, according to the organic light-emitting display apparatus of the current embodiment, because the third shield layer **SD3** exists between the data line **171** and the driving gate electrode **125a** of the driving TFT **T1**, the driving gate electrode **125a** of the driving TFT **T1** may not be affected or may be less affected by the data line **171**, and thus the organic light-emitting display apparatus may be able to display an image having a more accurate luminance and a relatively higher quality. For example, if the third shield layer **SD3** is a part of the second storage capacitor plate **127**, the second storage capacitor plate **127** is coupled to the driving voltage line **172** always having uniform electric potential, through the contact hole **168**, and thus the third shield layer **SD3** may also always have a uniform electric potential. Accordingly, an effect of an adjacent electric signal on the driving gate electrode **125a** of the driving TFT **T1** may be reduced.

(80) Of course, the third shield layer **SD3** may not only be disposed between the data line **171** and the driving gate electrode **125a**, and may also extend below the data line **171** as shown in FIG. **10**. Accordingly, the driving gate electrode **125a** of the driving TFT **T1** may be further shielded.

(81) Hereinabove, the organic light-emitting display apparatus may include the first shield layer **SD1**, the second shield layer **SD2**, and the third shield layer **SD3**, but alternatively, the organic light-emitting display apparatus may include only some of the first through third shield layers **SD1** through **SD3**. In other words, the organic light-emitting display apparatus may include at least any one of the first through third shield layers **SD1** through **SD3**.

(82) In the above embodiments, the compensating TFT **T3** and the initialization TFT **T4** include a dual gate electrode. However, an embodiment is not limited thereto and the organic light-emitting display apparatus may include the first or second shield layer **SD1** or **SD2** disposed between the data line **171** and a part of the compensating TFT **T3** and/or the initialization TFT **T4**, even if the compensating TFT **T3** and the initialization TFT **T4** include a single gate electrode.

(83) Meanwhile, all of the first through third shield layers **SD1** through **SD3** are included in the second gate wire as shown in FIGS. **2** and **5**, but an embodiment is not limited thereto. In other words, the first through third shield layers **SD1** through **SD3** may be a part of the second storage capacitor plate **127** or a part of the initialization voltage line **124**.

(84) FIG. **11** is a schematic diagram showing locations of a plurality of TFTs and a capacitor in a (sub) pixel of an organic light-emitting display apparatus, according to another embodiment of the present invention, and FIG. **12** is a cross-sectional view taken along the line XII-XII of FIG. **11**. Differences between the organic light-emitting display apparatuses of FIGS. **2** and **11** are shapes of the previous scan lines **122**, the initialization voltage lines **124**, and the initialization TFTs **T4**.

(85) Referring to FIGS. **11** and **12**, the initialization voltage line may be arranged in the same layer as the second storage capacitor plate **127**, or in the same layer as a pixel electrode. The initialization voltage line may be coupled to the initialization source electrode **176d** of the initialization TFT **T4** through the contact hole **162**. As described above with reference to FIG. **2**, the initialization drain electrode **177d** of the initialization TFT **T4** is electrically coupled to the compensating drain electrode **177c** of the compensating TFT **T3** and the driving gate electrode **125a** of the driving TFT **T1**.

(86) The previous scan line **122** that may be arranged in the same layer as the driving gate electrode **125a**, the scan line **121**, and the emission control line **123** may include two protrusions corresponding to a location of the initialization TFT **T4**. Here, the two protrusions may be the first and second gate electrodes **125d1** and **125d2** of the initialization TFT **T4**. At least a part of the second gate electrode **125d2** may be the second shield layer **SD2**.

(87) A dual gate electrode may have two parts overlapping a semiconductor layer. For example, referring to FIG. 11, the second gate electrode **125d2** of the initialization TFT **T4** may be a part of the previous scan line **122** extending along an x-axis without having to protrude from the previous scan line **122**, and a part **125d2'** of the first gate electrode **125d1** in the -x direction, which crosses a semiconductor layer near the initialization source electrode **176d**, may operate as a second gate electrode. However, in this case, a part between a part of the semiconductor layer corresponding to the part **125d2'** and a part of the semiconductor layer corresponding to the first gate electrode **125d1** is arranged adjacent to the data line **171** and is not shielded, and thus may be affected by the data line **171**.

(88) However, according to the organic light-emitting display apparatus of the some embodiments, the previous scan line **122** includes the two protrusions, wherein one of the protrusions operates as the first gate electrode **125d1** and the other one of the protrusions protrudes from the part **125d2'** of the previous scan line **122** and operates as the second gate electrode **125d2**. Here, the second gate electrode **125d2** shields the part between the part of the semiconductor layer corresponding to the part **125d2'** and the part of the semiconductor layer corresponding to the first gate electrode **125d1** from the data line **171**, and thus an unintended effect on the initialization TFT **T4** from the data line **171** may be effectively blocked or reduced.

(89) According to the initialization TFT **T4** having such a structure, the initialization TFT **T4** includes the first and second gate electrodes **125d1** and **125d2**, and one of the first and second gate electrodes **125d1** and **125d2** is at least partially disposed between the data line **171** and the semiconductor layer **131d2** that is a part between the source electrode **176d** and the other of the first and second gate electrodes **125d1** and **125d2** of the initialization TFT **T4**. In FIGS. 11 and 12, the second gate electrode **125d2** is at least partially disposed between the data line **171** and the semiconductor layer **131d2** that is the part between the source electrode **176d** and the first gate electrode **125d1** of the initialization TFT **T4**, and thus the semiconductor layer **131d2** is shielded from the data line **171**. In other words, the second gate electrode **125d2** is shown to be the second shield layer **SD2**. Here, the second gate electrode **125d2** may not only be arranged between the data line **171** and the semiconductor layer **131d2** as shown in FIGS. 11 and 12, but may also extend below the data line **171** in the -x direction. In FIG. 12, the data line **171** is arranged above the second gate electrode **125d2**, but if the data line **171** is arranged below the semiconductor layer **131d2** and the second gate electrode **125d2** is arranged between the data line **171** and the semiconductor layer **131d2**, the second gate electrode **125d2** may extend above the data line **171**.

(90) As such, the second shield layer **SD2** may be formed as the second gate wire as described above with reference to FIGS. 2, 5, and 9, but may alternatively be formed as the first gate wire as described with reference to FIGS. 11 and 12. If the first or third shield layer **SD1** or **SD3** is included as well as the second shield layer **SD2**, the first or third shield layer **SD1** or **SD3** may be formed as the first gate wire. In this case, the first or third shield layer **SD1** or **SD3** may not be electrically coupled to the second storage capacitor plate **127**, but may have an island shape and electrically float.

(91) Hereinabove, it is described that the parts of the driving TFT **T1**, the compensating TFT **T3**, and the initialization TFT **T4** are shielded from the data line **171**, but an embodiment is not limited thereto. In other words, if a TFT of a (sub) pixel of an organic light-emitting display apparatus is near the data line **171**, a shield layer may be disposed between the data line **171** and at least a part of the TFT such that the organic light-emitting display apparatus displays an image having high quality. The shield layer may be arranged at least one of between the data line **171** and a source electrode of the TFT, between the data line **171** and a drain electrode of the TFT, and between the data line **171** and a gate electrode of the TFT.

(92) Meanwhile, hereinabove, it is described that a shield layer is arranged between a data line and a part of a TFT, but an embodiment is not limited thereto. For example, an organic light-emitting display apparatus may include a TFT that includes a source electrode, a drain electrode, and a gate

electrode, a control signal line that is arranged in a layer different from the source electrode, the drain electrode, and the gate electrode and transmits a control signal, and a shield layer that is arranged between the control signal line and at least a part of the TFT. Here, the control signal line may be at least any one of the plurality of signal lines described above. In other words, the control signal line may be the scan line **121**, the previous scan line **122**, the emission control line **123**, the data line **171**, the driving voltage line **172**, or the initialization voltage line **124**. The shield layer may shield the at least the part of the TFT from the control signal line so as to block or reduce an effect of a control signal transmitted from the control signal line on the TFT.

(93) An embodiment of the present is not limited to an organic light-emitting display apparatus. An image having high quality may be displayed as long as a display apparatus including a TFT and a data line in a (sub) pixel has a shield layer in the same or similar manner described above.

(94) As described above, according to one or more embodiments of the present invention, a display apparatus capable of preventing quality deterioration of a displayed image may be realized.

(95) While one or more embodiments of the present invention have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present invention as defined by the following claims, and their equivalents.

Claims

1. A display apparatus comprising: a first transistor comprising a first gate electrode and a first semiconductor layer; a second transistor comprising a second gate electrode and a second semiconductor layer, the second semiconductor layer comprising a drain electrode and a source electrode; a connecting unit connected to the first gate electrode and the second semiconductor layer; and an electrode layer in a layer between the first semiconductor layer and the connecting unit, wherein the second semiconductor layer has a curved portion between the drain electrode and the source electrode of the second semiconductor layer, and wherein the electrode layer overlaps the curved portion of the second semiconductor layer.
2. The display apparatus of claim 1, wherein the electrode layer overlaps the first gate electrode and a part of the connecting unit in a plan view.
3. The display apparatus of claim 1, wherein the second gate electrode comprises two gate electrodes in a same layer.
4. The display apparatus of claim 3, wherein the curved portion of the second semiconductor layer is between the two gate electrodes of the second transistor in a plan view.
5. The display apparatus of claim 1, further comprising: a third transistor comprising a third gate electrode and a third semiconductor layer; and a shield layer in a layer between the third semiconductor layer and the connecting unit, wherein the connecting unit is connected to the third semiconductor layer, wherein the third semiconductor layer has a curved portion, and wherein the shield layer overlaps the curved portion of the third semiconductor layer.
6. The display apparatus of claim 5, wherein the third gate electrode comprises two gate electrodes in a same layer.
7. The display apparatus of claim 6, wherein the curved portion of the third semiconductor layer is between the two gate electrodes of the third transistor in a plan view.
8. The display apparatus of claim 1, further comprising: a third transistor comprising a third gate electrode and a third semiconductor layer; and an initialization voltage line in a layer between the third semiconductor layer and the connecting unit, and connected to the third semiconductor layer, wherein the connecting unit extends in a first direction and the initialization voltage line extends in a second direction different from the first direction.
9. The display apparatus of claim 8, wherein the third semiconductor layer has a curved portion, and wherein the initialization voltage line overlaps the curved portion of the third semiconductor

layer.

10. The display apparatus of claim 1, further comprising a driving voltage line extending in a first direction and connected to the electrode layer.

11. The display apparatus of claim 10, wherein the driving voltage line and the connecting unit are in a same layer.

12. The display apparatus of claim 1, further comprising: a fourth transistor comprising a fourth gate electrode and a fourth semiconductor layer; and a data line connected to the fourth semiconductor layer, wherein the data line and the connecting unit are in a same layer.

13. The display apparatus of claim 12, wherein the data line extends in a first direction and overlaps a part of the electrode layer.

14. The display apparatus of claim 13, wherein a length along the first direction of the overlapping part of the electrode layer with the data line is longer than a length along the first direction of the first gate electrode.

15. The display apparatus of claim 12, further comprising a driving voltage line extending in a first direction and connected to the electrode layer, wherein the driving voltage line and the data line are in a same layer and spaced in parallel.

16. The display apparatus of claim 15, wherein the driving voltage line is between the data line and the connecting unit, in a plan view.

17. The display apparatus of claim 1, wherein the first semiconductor layer has a curved portion, and wherein the first gate electrode and the electrode layer overlaps the curved portion of the first semiconductor layer.

18. The display apparatus of claim 1, further comprising a capacitor comprising a first electrode and a second electrode, wherein the first electrode comprises the first gate electrode and the second electrode comprises the electrode layer.

19. The display apparatus of claim 1, wherein the first semiconductor layer and the second semiconductor layer comprise polysilicon.
